Metabolomics (2016) 12:71
DOI 10.1007/s11306-016-0987-8

ORIGINAL ARTICLE

Metabolomic signatures of increases in temperature and ocean acidification from the reef-building coral, *Pocillopora damicornis*

Emilia M. Sogin[1] · Hollie M. Putnam[1] · Paul E. Anderson[2] · Ruth D. Gates[1]

Received: 6 August 2015 / Accepted: 17 December 2015 / Published online: 8 March 2016
© Springer Science+Business Media New York 2016

Abstract
Introduction As a changing climate threatens the persistence of terrestrial and marine ecosystems by altering community composition and function, differential performance of taxa highlights the need for predictive metrics and mechanistic understanding of the factors underlying positive performance in the face of environmental disturbances. Biochemical reactions within cells provide a snapshot of molecular regulation and flexibility during exposure to environmental stressors. However, because the organism is the unit of selection there is a need for the integration of metabolite data with organism physiology to understand mechanisms responsible for individual success under a changing climate.
Objectives Our study aims to characterize the molecular response of reef corals to simulated global climate change stressors. Furthermore, we seek to relate changes in the molecular physiology to observations in overall colony response.
Methods To this end, we applied a non-targeted metabolomic approach to describe lipid and primary metabolite composition after exposure of the reef-building coral *Pocillopora damicornis* to ambient and elevated experimental climate change conditions. We compared these metabolite data to organism physiology, specifically the key processes of photosynthesis, respiration, and calcification.
Results Corals significantly altered their lipid and primary metabolite profiles in response to experimental treatments. Primary metabolite profiles predicted organisms' net photosynthesis, but not calcification or respiration measures. Despite challenges in metabolome annotation, our data indicated corals alter carbohydrate composition, cell structural lipids, and signaling compounds in response to elevated treatment conditions.
Conclusions The integration of metabolite and physiological data highlights the predictive power of metabolomics in defining organism performance and provides biomarkers for future studies. Here, we present a multivariate biomarker approach to assess climate change impacts and advance our mechanistic understanding of stress response in this keystone species.

Keywords Metabolomics · Lipidomics · *Pocillopora damicornis* · Ecological disturbance · Ocean acidification · Global climate change

1 Introduction

Changing climate, driven by increases in anthropogenic green house gas emissions, alters the trajectory of ocean ecosystems (Doney et al. 2009, 2012). For instance, chronic exposure to ocean acidification, or elevation in the partial pressure of carbon dioxide (pCO_2) resulting in a decrease in ocean pH, and elevated sea-surface temperatures negatively impacts reproduction, calcification, growth, and survival of marine organisms (Kroeker et al. 2010). In particular, increases in temperature and ocean

Electronic supplementary material The online version of this article (doi:10.1007/s11306-016-0987-8) contains supplementary material, which is available to authorized users.

✉ Emilia M. Sogin
esogin@hawaii.edu

[1] Hawaii Institute of Marine Biology, University of Hawaii at Manoa, 46-007 Lilipuna Road, Kaneohe, HI 96744, USA

[2] Department of Computer Science, College of Charleston, 66 George Street, Charleston 29424, SC, USA

acidification pose a significant and imminent threat to reef-building corals (Doney et al. 2012; Pandolfi et al. 2011), which form the structural foundation of tropical reef ecosystems that provide essential services (e.g., fisheries habitat, eco-tourism, costal protection) to millions of people (Hoegh-Guldberg et al. 2007; Moberg and Folke 1999).

Combined increases in temperature and ocean acidification during the past several decades have decreased coral cover by approximately 1–5 % per year in some regions (Bruno and Selig 2007, Gardner et al. 2003) representing coral cover loss as great as 50–80 % (De'ath et al. 2012; Gardner et al. 2003). This loss of reef building coral is propelled by declines in coral growth and increases in coral mortality (Hughes et al. 2007), and preceded by shifts in coral physiological performance. For example, abnormally high sea-surface temperatures lead to mass bleaching, or the expulsion or loss the corals' symbiotic algal partners (e.g., *Symbiodinium* spp.) causing colonies to pale in color (Glynn 1984; Brown 1997). Bleaching events include disruptions in algal photosynthesis, thereby limiting energy transfer to the coral, reducing coral growth (Goreau and Macfarlane 1990) and detrimentally impacting reproduction (Baird and Marshall 2002; Mendes and Woodley 2002). Additionally, increases in ocean acidification can limit the capacity of corals to produce calcium carbonate structures, leading to declines in reef accretion and reduced coral growth (Jury et al. 2010; Langdon and Atkinson 2005; Ries et al. 2009; Rodolfo-Metalpa et al. 2010; Putron et al. 2011). Lower ocean pH can also reduce algae photosynthetic productivity, increases coral bleaching, and depresses colony respiration (Anthony et al. 2008; Crawley et al. 2010; Kaniewska et al. 2012; Marubini et al. 2008). These physiological responses are consistent with shifts in gene expression patterns that indicate global climate change stressors induce metabolic suppression, increases oxidative stress, and symbiont expulsion rates (Kaniewska et al. 2012).

Environmental stressors are projected to intensify as anthropogenic activities continue to alter oceans, and therefore threaten the persistence of reef ecosystems (Doney et al. 2012; Pandolfi et al. 2011). Therefore, there is a pressing need to develop analytical tools that can rapidly measure changes in coral performance and investigate mechanisms underling coral responses to global climate change disturbances (Downs et al. 2005, 2012). One promising approach is to investigate molecular markers at the level of transcripts, proteins, and metabolites using 'omics' approaches (e.g., next generation sequencing and high throughput chemical profiling technologies). All organisms, including reef-building corals, interact with their environment by shifting gene and protein expression patterns (Kültz 2005) to facilitate homeostasis. Recent applications of 'omics' profiling tools have enhanced efforts to describe coral resistance, resilience, and function (Meyer and Weis 2012). While there are challenges and limitations in interpreting these large datasets (Meyer and Weis 2012), notably the lack of annotation and chemical references, the global analyses of genes (Shinzato et al. 2011), transcripts (Barshis et al. 2013), and proteins (Peng et al. 2012) are advancing the understanding of coral colony responses to ecological disturbances. For instance, experimental evidence suggests corals alter gene expression to compensate for high temperatures (DeSalvo et al. 2010) and to acclimatize to thermally variable reef habitats (Barshis et al. 2013). Simulated ocean acidification conditions induce shifts in the expression of genes involved specifically in metabolism and calcification pathways (Moya et al. 2012). Finally, such differential gene expression elicits dynamic cellular responses after stress exposure (Seneca and Palumbi 2015). However, gene and protein expression regulates the transcription of the gene sequence into mRNA and subsequent translation into protein, the dynamics of which can be somewhat disconnected from organism physiology. Furthermore, these techniques can be cost prohibitive for reef-wide monitoring programs and are not currently used by managers to assess long-term impacts of global climate change or other stressors on reef condition.

One underexplored avenue for beginning to describe cellular pathways implicated in coral stress responses is to analyze changes in metabolite composition. Metabolites are small compounds (e.g., glucose, glycerol, and alanine) that are products and intermediates of metabolism, and respond to shifts in the environment (Bundy et al. 2009). As such, metabolite profiles play essential roles in biochemical pathways that underlie individual phenotype (Michal and Schomburg 1999; Wagner et al. 2013) and, in eukarotic organisms, are impacted by the presence of associated microoganisms (e.g., Bouslimani et al. 2015; McHardy et al. 2013; Shankar et al. 2015). Similarly, reef-building coral metabolite profiles can originate from the coral host and its diverse symbiotic associations with microbial partners, including viruses, archaea, bacteria, dinoflagellates and fungi (Gates and Ainsworth 2011; Krediet et al. 2013; Rosenberg et al. 2007; Rowher et al. 2002; Thurber et al. 2008). Together, these symbiotic partners comprise the coral holobiont, where metabolites are the currency of the endosymbiosis (Gates and Ainsworth 2011; Gordon and Leggat 2010). For instance, corals form an intimate union with dinoflagellate algae partners in the genus *Symbiodinium*. In exchange for nitrogen compounds, the algae produce and translocate organic metabolites to the host, which can be used to meet up to 95 % of the coral's daily nutritional needs (Muscatine and Porter 1977). Algal photosynthate is primarily composed of glycerol with smaller quantities of glucose, amino acids, organic acids, and free fatty acids (Burriesci et al. 2012;

Muscatine and Cernichiari 1969; Papina et al. 2003; Rahav et al. 1989; Whitehead and Douglas 2003). These compounds support host nutritional pathways by providing energetic resources needed for growth, respiration, reproduction, and calcification processes (Davy et al. 2012; Goreau et al. 1971; von Holt 1968; Muscatine and Cernichiari 1969; Yellowlees et al. 2008). For instance, translocated compounds are stored in lipid droplets and starch granules, which can be remobilized when nutritional demands are not being met (Kopp et al. 2015). These photosynthetic products are also incorporated into the epithelial tissue layer, where the coral uses them for growth and cell maintenance (Kopp et al. 2015). Like *Symbiodinium*, prokaryotic partners are also involved in contributing to the bulk coral holobiont profiles. For example, coral-associated bacteria are involved in the cycling of sulfur compounds, such as the degradation of dimethylsulfoniopropionate (DMSP) to dimethyl sulfide (Raina et al. 2009), and the fixation of nitrogen compounds (Lesser et al. 2004). In addition to the associated microbial partners, corals themselves also produce and mobilize other small compounds, including steroids, DMSP, betaines, a diverse assemblage of lipids, and mycosporine-like amino acids (Banaszak et al. 2000, 2006; Dunn et al. 2012; Tapiolas et al. 2010; Tarrant et al. 1999, 2003; Yancey et al. 2010). While the diversity of metabolites plays an essential role in organism function, few studies (but see Dunn et al. 2012; Sogin et al. 2014; Klueter et al. 2015) have attempted to apply non-targeted metabolomic approaches to describe the presence and abundance of these compounds in reef-building corals.

An organism's metabolite composition links its genome to its phenome (Fiehn 2001) through the environmental filter. Non-targeted metabolomics approaches can be used to quickly assess many metabolites simultaneously to provide a more complete picture of coral metabolic status (e.g., Burriesci et al. 2012; Dunn et al. 2012; Sogin et al. 2014). In a marine setting, researchers have used metabolomics to study the response of bivalves (Ellis et al. 2014), crustaceans (Hammer et al. 2012), and marine microbial communities (Coelho et al. 2015) to simulated future levels of high temperature and ocean acidification. These unbiased methods allow for the development of a mechanistic understanding of individual response to stress through evaluation of molecular pathways.

In this study, we describe changes in the metabolite profiles of the corals, including their symbiotic partners, as a function of simulated global climate change stressors. Furthermore, we characterize the relationship between metabolite signatures and coral physiology. Due to the lack of capacity to physically separate the coral host from its symbionts without potentially degrading the tissue and the limitations of the reference databases for species-specific metabolites, it was beyond the scope of this study to identify exact contributions of the coral and its microbial partners to the metabolite profiles. However, the described approach enables us to (1) identify shifts in metabolite profiles that respond to simulated future ocean conditions, (2) assess the predictive capacity of the metabolite profiles for coral physiology, and (3) hypothesize possible mechanisms underlying coral responses to simulated global climate change stressors.

2 Materials and methods

2.1 Experimental exposure and coral sampling

The Indo-Pacific reef coral *Pocillopora damicornis* (Linnaeus 1758) was sampled for metabolomic analysis from a manipulative experiment described in Putnam and Gates (2015). In brief, adult colonies (n = 11 per treatment) were exposed to either ambient (26.5 °C and 416 µatm pCO_2) or high (29 °C and 805 µatm pCO_2) synergistic levels of temperature and pCO_2 for 1.5 months. The combination of factors applied in the high treatment conditions represents a future scenario of simultaneous increases in sea-surface temperatures and ocean acidification. The 2.5 °C increase in temperature and a doubling of pCO_2 are in line with the expectations of RCP 6.0 by the year 2100 (IPCC 2014). These are conservative and realistic scenarios for Kaneohe Bay, HI as water temperatures have exceeded 31 °C during 2014 and 2015 bleaching events (NOAA http://tidesandcurrents.noaa.gov/physocean.html?id=1612480) and pCO_2 can reach >600 µatm under normal conditions in the Bay (Drupp et al. 2011). Following experimental manipulation, coral fragments were snap frozen in liquid nitrogen to halt metabolism and processed by the University of California Davis West Coast Metabolomics core facility for metabolome analysis. To facilitate broader coverage of the coral metabolome, non-polar and polar compounds were targeted for mass spectrometry (MS) analysis using lipid and primary metabolite analytical platforms, respectively.

2.2 Lipid profiling with liquid chromatography-tandem mass spectrometry (LC–MS)

Coral lipids were extracted following the Matyash protocol (Matyash et al. 2008), which limits contamination of the non-polar phase by proteins and polar compounds from 40 mg of a pulverized coral fragment (i.e., coral animal and its eukaryotic and prokaryotic symbionts). Liquid chromatography–tandem MS (LC–MS) profiles were obtained from the organic phase, which was dried and reconstituted in 65 µL of methanol:toluene (9:1, v/v)

containing a 12-[[(cyclohexylamino)carbonyl]amino]-dodecanoic acid internal standard. Sample aliquots (1.67 µL for positive and 5 µL for negative ionization modes) were eluted at 0.6 mL/min from an Acquity charged surface hybrid (CHS) C18 column (Waters; 100 mm × 2.1 mm; 1.7 µm particle) using an Agilent 1290 (Agilent Technologies, Santa Clara, CA, USA) ultra-high pressure LC system. Compounds were separated using a gradient composed of (A) acetonitrile:water (60:40, v/v) and (B) isopropanol:acetonitrile (90:10 v/v) both containing 10 mM ammonium formate and 0.1 % formic acid. Gradient conditions were: 15–30 % B 0–2 min, 30–45 % B 2.0–2.5 min, 45–82 % B 2.5–11 min, 82–99 % B 11–12 min, then returned to starting conditions at 12.1 min followed by equilibration at 15 % B 12.1–15 min. Electrospray ionization (ESI) was used to ionize column elutants in both positive and negative modes using an Agilent 6530 accurate-mass quadrupole time of flight (Q-TOF) LC–MS and an Agilent 6550 iFunnel Q-TOF, respectively. The capillary voltage was set to +3.5 and −3.5 kV, and the collision energy to 25 or 40 eV for positive and negative modes. Mass-to-charge ratios (*m/z*) were scanned from 60 to 1700 Da and spectra acquired every 2 s. MassHunter (Qual v. B05.00) was used to find peaks from the raw data in up to 300 chromatograms. These peaks were imported into Mass Profiler Professional for alignment to determine which peaks occur in at least 30 % of the chromatograms. These peaks were then quantified with MassHunter. Resulting accurate mass data and tandem MS/MS spectra were compared to LipidBlast libraries for compound identification (Kind et al. 2013). Lipid peak heights were normalized to total average metabolite content across all samples (Fiehn et al. 2008). Resulting metabolite data were imported into R (version 3.0.0, R Core Development Team) for statistical analysis.

2.3 Primary metabolite profiling with gas chromatography-mass spectrometry (GC–MS)

Coral primary metabolites were extracted and derivatized following methods from (Kind and Fiehn 2006). Briefly, 40 mg of pulverized coral nubbin (i.e., coral animal and its eukaryotic and prokaryotic symbionts) was extracted in 1 mL of acetonitrile:isoproponal:water (3:3:2; v/v/v) at 4 °C for 5 min. Extracts were centrifuged (2 min at 14,000 rcf) and the supernatant was evaporated to dryness in a cold trap concentrator prior to derivatization. Analytes were separated on a Rtx-5Sil column (30 m × 0.25 mm; 0.25 µm Rxi-5Sil) attached to an Agilent 6890 GC and ionized on a Leco Pegasus IV mass spectrometer (St Joseph, MI, USA) following acquisition parameters previously reported (Fiehn et al. 2008). Resulting metabolite profiles were processed using UC Davis' BinBase workflow, which preforms data processing including peak detection at signal/noise levels of 5:1 throughout the chromatogram. Resulting apex masses are reported for use in the BinBase algorithm to facilitate metabolite identification and quantification (Fiehn et al. 2005). Metabolites were identified through comparison to the BinBase database (http://fiehnlab.ucdavis.edu/Metabolite-Library-2007) and peak heights were normalized to total metabolite content (Fiehn et al. 2008). Resulting data were imported into R for statistical analysis.

2.4 Statistical analysis

Relative standard deviations (% RSD = mean/standard deviation × 100) assess variation in metabolite composition among individuals (Parsons et al. 2009). To determine if variability in MS profiles differed between *P. damicornis* corals exposed to high and ambient conditions, RSD values were calculated across all metabolites for both experimental treatments. A Kruskal–Wallis test statistically compared RSD values between treatment groups.

To facilitate comparison of metabolite composition among treatment groups, normalized compound abundances were auto-scaled (mean centered and scaled to unity) prior to multivariate analysis (van Den Berg et al. 2006). An orthogonal partial least squares-discriminate analysis (OPLS-DA) was used to model the relationship between *P. damicornis* metabolite composition and experimental treatments (for source code see https://github.com/Anderson-Lab/OPLS). Model accuracy was calculated using a tenfold cross validation technique (Zhang 1993) to test the capacity for metabolite abundance to predict coral experimental exposure. The resulting goodness-of-fit parameter, Q^2, was compared to its permutated null distribution to test for model significance ($p < 0.05$).

To identify metabolites driving separation in the OPLS-DA models, variable loading scores were also compared to their permuted null distributions. Metabolites located in the distribution tails ($\alpha < 0.01$) contributed significantly to separation in the model. To further investigate significant metabolites, percent differences in compound abundances were calculated in individuals exposed to high conditions in comparison to their ambient counterparts. Identified metabolites were compared to the Kyoto Encyclopedia of Genes and Genomes (KEGG) and the Human Metabolome Database (HMDB) to annotate metabolite function.

To investigate whether metabolites correlate to variation in coral physiological parameters, OPLS regression models were created to relate primary metabolite and lipid data to previously collected net photosynthesis, respiration, and calcification results (measured from the same coral individuals) reported in Putnam and Gates (2015). OPLS regression models remove uncorrelated variation in the Y

response variables (t-orthogonal) to facilitate model interpretation (Trygg and Wold 2002). Again, Q^2 was calculated using tenfold cross validation and compared to its null distribution to test for model significance. In this setting, Q^2 describes the total amount of variation captured by the model. For significant models, OPLS variable loadings were compared to their permuted null distributions to determine if the metabolites they represent contribute significantly ($\alpha < 0.01$) to the measured relationship. Those metabolites found to influence the model were further investigated by calculating Pearson's R^2 correlations between metabolite abundances and physiological variables.

3 Results and discussion

Chronic exposure to ocean acidification and elevated sea-surface temperatures negatively impacts the health of many marine organisms (Kroeker et al. 2010). Shifts in ocean conditions associated with a warming climate pose a significant threat to reef-building corals (Hoegh-Guldberg et al. 2007; Pandolfi et al. 2011). However, traditional monitoring methods, such as tracking changes in coral reef cover and colony pigmentation, are limited in their capacity to proactively measure changes in coral health at the molecular level. Here, the use of primary metabolite profiling allowed us to predict net photosynthetic performance of the coral, and with the addition of lipidomics analysis begin to elucidate the molecular mechanisms of response of *P. damicornis* to climate change stressors. These metabolite profiles represent the metabolic contribution of all coral associated symbiotic partners, including the diverse assemblage of viruses, bacteria, Archaea, fungi and dinoflagellates (Gates and Ainsworth 2011; Gordon and Leggat 2010).

3.1 Primary metabolite and lipid profiles as indicators of global climate change stress

In total, MS methods detected 1058 unique masses, consisting of 450 lipids and 608 primary metabolites. The vast majority (79 %) of profiled compounds did not match LipidBlast or BinBase records (Fig. 1; Table S1). KEGG search results (Kanehisa and Goto 2000) provided functional annotations of identified compounds (Table S2).

Reef-building corals are dynamic organisms, exhibiting significant diversity in colony morphology, physiology, microbiomes, and ecological tolerance within and among species (Ainsworth et al. 2010; Loya 2001; Yost et al. 2013). This variation is reflected in the response of lipid and primary metabolite profiles to treatment conditions. RSD scores were significantly lower (p value < 0.05) in corals exposed to

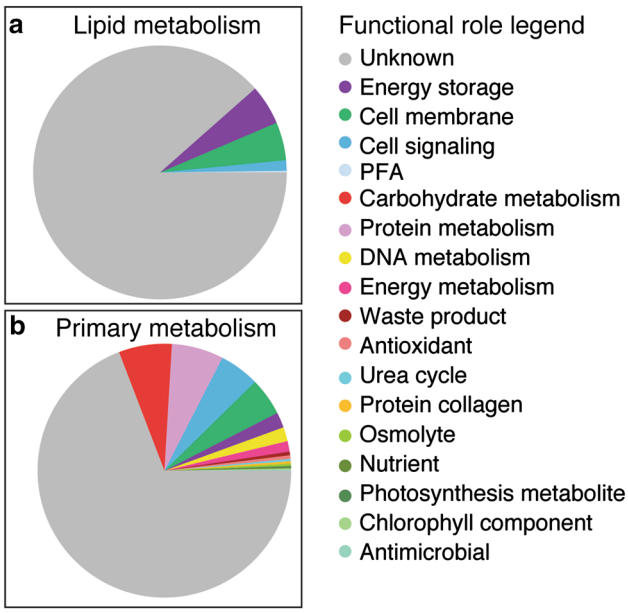

Fig. 1 Coral metabolite functions. *Pie* charts showing the proportion of metabolites falling into functional categories for compounds measured by **a** lipid metabolism and **b** primary metabolism platforms (Color figure online)

elevated conditions (median % $RSD_{lipid} = 81.9$, median % $RSD_{primary} = 69.4$) in comparison to their ambient (median % $RSD_{lipid} = 56.9$, median % $RSD_{primary} = 55$) counterparts (Fig. 2a, c).

Metabolites shift in specific and predicable patterns after exposure to factors such as changing environmental conditions associated with global climate change (Dickinson et al. 2012), disease (Viant et al. 2003), and microbial communities (Wikoff et al. 2009). These patterns are consistent in reef-building corals; both lipid and primary metabolite profiles of *P. damicornis* differed significantly (p-value < 0.05, Fig. 2b, d) between the experimental treatments, as determined by the OPLS-DA model. Our OPLS-DA approach allows the variation in the metabolomic data to be partitioned between (t_1 axis) and within (t_{O1} axis) treatment groups. This allows one to test that hypothesis that shifts in coral metabolite profiles can predict initial experimental treatment conditions. Both OPLS-DA models (Fig. 2b, d) indicate the measured primary metabolite and lipid profiles accurately predict coral exposed to ambient or high experimental treatments. When comparing the OPLS-DA model output, it is clear primary metabolite profiles are better indicators of experimental treatment exposure since there is complete separation of the experimental groups along the t_1-axis (non-overlapping, almost vertical lines of data points), above and beyond the variation within samples (t_{O1} axis). In support of these patterns, the magnitude of Q^2 in the lipid OPLS-DA model ($Q^2 = 0.198$) was only 60 % of the primary

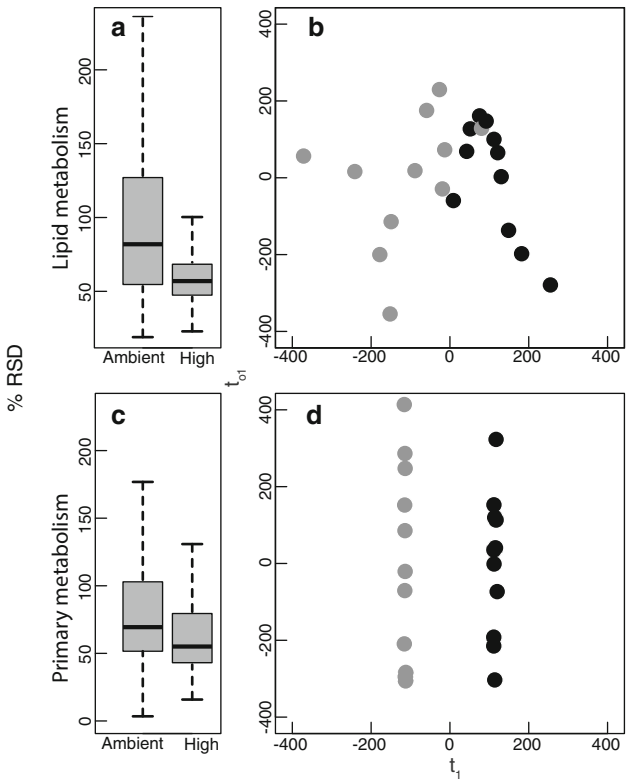

Fig. 2 *Pocillopora damicornis* metabolite profiles change with exposure to global climate change stressors. **a**, **c** Relative standard deviation *box-plots* comparing variation in metabolite abundances between treatment groups for **a** lipid and **c** primary metabolite profiling platforms. The median is indicated (*black bar*) along with the quartile ranges. **b**, **d** OPLS-DA models comparing coral metabolomes after exposure to ambient (*gray circles*) and high (*black circles*) treatment conditions for data collected by **b** lipid and **d** primary metabolite profiling platforms. Separation along the t (t_1) and t-orthogonal (t_{o1}) axis represents variation within and between treatments. For metabolites driving separation between treatments, see Table S2

metabolite OPLS-DA model ($Q^2 = 0.321$). This suggests that measured primary metabolite profiles are better predictors of global climate change exposure than lipid compounds.

The primary metabolite profiles show strong and uniform discrimination between corals exposed to ambient and future ocean conditions. This is indicated by the clear separation of coral samples exposed to high and ambient treatments (Fig. 2d). Consequently, *P. damicornis* metabolic response to climate change stressors at the primary metabolite level is consistent among individuals, suggesting a focused cellular metabolic reaction within a population (Parsons et al. 2009). Our observations are as expected, as many of the measured primary metabolites are involved in symbiotic energy production, recycling, and catabolism (e.g., carbohydrates; Gordon and Leggat 2010; Muscatine 1967; Von Holt 1968; Yellowlees et al. 2008). Changing climatic conditions negatively impact the coral-algal nutritional symbiosis (Anthony et al. 2008; Crawley et al. 2010; Lesser 2011), likely resulting in loss of translocated photosynthate with downstream metabolic consequences.

Discrimination of coral samples based on lipid profiles was less uniform and overall produced a weaker model than that for primary metabolites (Fig. 2b). This result suggests corals shift primary metabolites in more predictable patterns in comparison to lipid compounds when responding to simulated stress events. While there is a clear need for additional dynamic studies to characterize this relationship, the observations made here may be related to differences in the turnover rates of small compounds, where lipid pools may be more energetically costly to regulate then primary metabolites. Major lipid regulation in corals typically only occurs after exposure to extreme stress events (Fitt et al. 2009; Grottoli et al. 2004). Therefore, it is possible our simulated disturbance event was not of the magnitude necessary to elicit uniform changes in lipid content across individuals. For instance in plants, leaf membranes primarily composed of lipids are resistant to drought stress until the event reaches a threshold severe enough to induce remodeling (Gigon et al. 2004). The compounds detected by our lipid-profiling platform are involved in maintaining cell structures, facilitating signaling cascades, and supplying raw material for storage reservoirs (Fig. 3; Table S3). Changes in lipid bilayer membranes can disrupt chemical gradients needed for energy production and lead to cell death (van Meer et al. 2008). Consequently, many organisms actively regulate other small compounds, such as osmoprotectants or compatible solutes, to protect cellular structures and membranes from degradation (Yancey 2005).

The promise of applying metabolomics to improve understanding of the biology of ecologically relevant organisms is great (Bundy et al. 2009), but is, at this time, constrained by the lack of species-specific database tools, which limits metabolite identification. In this study for example, 90 and 62 % of the metabolites driving separation in lipid and primary metabolism profiles (Fig. 3a, b) observed in the OPLS-DA analysis (Fig. 2b, d) do not match the database records (e.g., BinBase or LipidBlast). As such, unknown compounds play a major role in describing changes in coral metabolite composition after prolonged exposure to high temperature and pCO_2 conditions. Conversely, twenty compounds driving separation between corals exposed to experimental treatments (Figs. 2 b, d, 3; Table S3) matched entries in existing databases. The functional analysis of these compounds reveals that *P. damicornis* shifts carbohydrate pools, lipid composition, and cell signaling compounds in response to elevated treatment conditions.

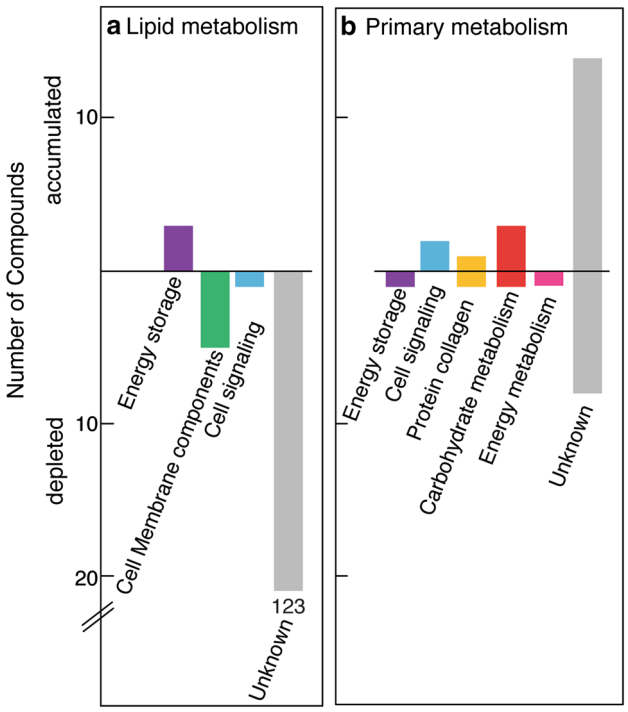

Fig. 3 Known functional roles of compounds describing metabolic differences between experimental treatments. *Bar charts* showing the number of metabolites accumulated and depleted in corals exposed to high treatment conditions relative to their ambient counterparts. Only compounds significant in the OPLS-DA analyses are represented for both the **a** LC–MS and **b** GC–TOF–MS datasets. Metabolites are represented once for each known functional role (Color figure online)

Investigation of the metabolites responsible for the discrimination of samples between the two treatments shows that corals are regulating carbohydrates and metabolites involved in cell structure and cell signaling pathways. This information is critical to describe mechanisms underlying coral response to long-term climate change stressors. Indeed, patterns in metabolite regulation are consistent across eukaryotic taxa, suggesting our metabolomic profiling approaches are capturing conserved organismal responses to disturbance (Table S4 and references therein). For instance, carbohydrate reprogramming may compensate for reduction in nutritional input to the holobiont by activating central metabolism pathways, as observed in the plant temperature metabolome, in an attempt to maintain physiological activity (Guy et al. 2008). Alternatively, changes in sugar chemistry could be combating osmotic stress (Kaplan et al. 2007; Guy et al. 2008). We expect both of these responses to occur in a coral experiencing a bleaching response, as observed in the high-treatment samples measured by a decline in photosynthesis (Putnam and Gates 2015). Despite a weaker discrimination model from lipid profiles, a decrease in phosphatidylcholines (PC), which are a major component of cell membranes (van Meer et al. 2008) and serve as structural lipids in corals (Patton et al. 1977), suggests exposure to elevated treatment conditions are remodeling coral lipids that contribute to the structural integrity of lipid bi-layers and cell membranes. Furthermore, we observe an increase in trans-4-hydroxyproline with a parallel decrease in 4-hydroxyproline, which is indicative of protein collagen depletion or turnover of damaged proteins (Bolling and Fiehn 2005). Alternatively, alterations in coral lipids may also be related to shifts in photosynthetic activity, as light driven photosynthesis is coupled to the biosynthesis of phospholipids (Oku et al. 2003; Patton et al. 1977). Finally, increases in cell signaling compounds indicate corals or their symbionts activate metabolic pathways to meet cellular homeostatic requirements.

The data does not allow us to conclusively identify all the processes corals regulate upon exposure due to the large number of unknown compounds (both lipid and primary metabolites) detected (79 %). Given metabolomic techniques have only recently been developed and applied to the study of reef corals (but see Burriesci et al. 2012; Dunn et al. 2012; Gordon et al. 2013; Sogin et al. 2014), the large number of un-annotated compounds is not surprising. Environmental metabolomics studies using non-model biological systems chronically suffer from decreased capacity to match small compounds to database entries (e.g., Aslund et al. 2011; Lardon et al. 2013; Schock et al. 2010). This is linked to a lack of non-mammalian metabolite databases (Bundy et al. 2009). Expansion of databases and targeted metabolite tracking will further the mechanistic capacity of metabolomic approaches.

3.2 Coral metabolomic profiles can predict net photosynthesis

To determine if coral metabolomic profiles reflect coral physiology, OPLS regression modeled the relationship between coral physiology parameters (respiration, calcification and net photosynthesis) reported in Putnam and Gates (2015) and lipid and primary metabolite profiles (Table S5). Only the model between primary metabolite profiles and net photosynthesis was significant ($Q^2 = 0.36$, $p < 0.01$; Table S6), with metabolite profiles predicting 36 % of the observed variation in net photosynthesis (Fig. 4). Six of the 35 metabolites that significantly contributed to this response were identified based on matches to the BinBase database. These represent compounds involved in protein collagen production, cell signaling, antioxidant production, protein synthesis, and carbohydrate metabolism (Table S6).

Shifts in coral primary metabolite pools predict net photosynthetic values (Fig. 4, Table S5), indicating there is a direct relationship between detected compounds and coral photosynthesis. However, we did not detect a

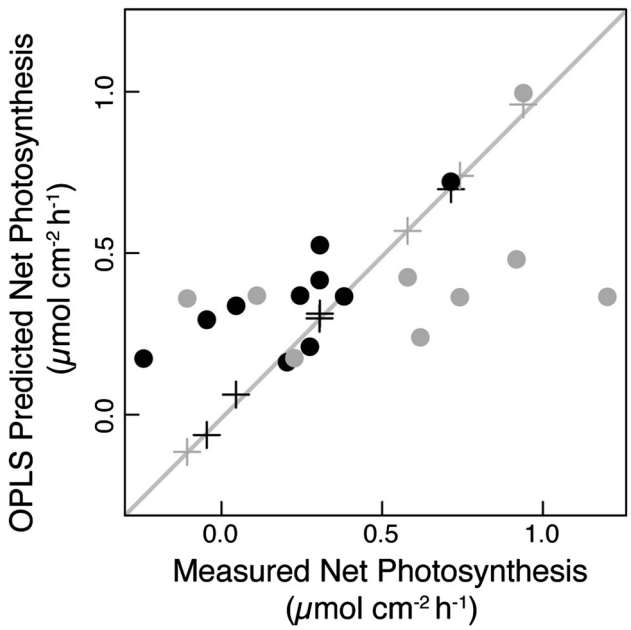

Fig. 4 Coral primary metabolite profiles are correlated to colony net photosynthesis. Predicted net photosynthetic values (from Putnam and Gates 2015) from metabolite composition in comparison to measured values. The *gray line* represents a 1:1 line. Values from *Pocillopora damicornis* samples originating from ambient and high treatment conditions are represented by *gray* and *black symbols* respectively. *Circles* and *crosses* represent the OPLS regression analysis validation and test data results, respectively. As indicated by the OPLS regression model, primary metabolite profiles describe 36 % of the variation in measured net photosynthesis ($Q^2 = 0.389$, $p < 0.02$). For OPLS regression results and metabolites contributing to this response, see Tables S3 and S4

relationship between primary metabolites or lipids and coral respiration, or calcification responses (Table S5). This suggests the products related to algal photosynthesis dominate primary metabolite pools. Algal photosynthesis drives the production of organic carbon compounds, which are catabolized to produce metabolites involved in major metabolic pathways (Stitt et al. 2010). Plants and algae achieve cellular homeostasis by regulating primary metabolism genes (Dittami et al. 2009) and reprogramming carbohydrate chemistry to compensate for reduction in photosynthetic activity (Kaplan et al. 2004). Considering *Symbiodinium* cells are densely packed into the coral gastrodermal tissues, individual metabolite profiles are likely comprised of products and intermediates tied to algal photosynthate. However, lacking tracer-based studies, it is challenging to identify which compounds are results of photosynthetic activity of the endosymbionts. Even so, our regression approach identified specific metabolites and pathways (Table S6) relating to shifts in net photosynthesis.

The lack of a relationship between the metabolite and lipid data to coral respiration and calcification (Table S5) indicates we are not capturing the products and intermediates of these physiological responses. Recalculation of the OPLS model using only detected metabolites that contribute to respiration (i.e., glycolysis, Kreb's cycle) was still unable to predict physiology values from metabolite composition (data not shown). This null result suggests that our approach is not measuring metabolites directly related to respiration, or more likely that the compounds driving this response are unknowns. To better characterize metabolites linked to host respiration pathways, a logical next step is to conclusively identify primary metabolites and lipids that are being produced and consumed by the coral. While we are unable to do this from the bulk coral metabolite profiles described, recently optimized data collection techniques that aim to separate the host and symbiont fractions of the coral holobiont could be used in future studies (see Burriesci et al. 2012), which may facilitate the integration of physiological and metabolomics data. Furthermore, because calcification, respiration and photosynthesis pathways are interrelated (Al-Horani et al. 2003; Allemand et al. 2004; Gattuso et al. 1999), and likely share metabolite endpoints, signals involved in calcification processes maybe swamped by noise generated from other physiological processes. To begin to actively characterize organic molecules that help facilitate calcification in reef corals, future experiments should measure metabolite profiles directly from the aboral tissue in contact with the skeleton and organic matrix by removing oral coral and symbiont tissue layers. Based on past efforts demonstrating the link between coral calcification and the organic matrix composition (Allemand et al. 1998; Constantz and Weiner 1988; Ramos-Silva et al. 2014), we expect organic matrix metabolites to be strongly associated with calcification. This approach may also allow researchers to begin to describe unknown compounds that help corals produce calcium carbonate skeletons. There is a clear need to develop a targeted profiling approach to describe metabolites underlying these responses.

By measuring coral metabolites we can use minimally invasive coral biopsies and rapid sample data acquisition to capture dynamic changes in colony primary production. Coral net photosynthesis is positively correlated to 5-deoxy-5-(methylthio) adenosine, 3-methoxytyrosine, and threonic acid. These metabolites contribute to energy and carbohydrate metabolism pathways and antioxidant production, all of which are byproducts of photosynthetic reactions. Net photosynthesis is also negatively correlated to the accumulation of trans-4-hydroxyproline, 2-phenylpropanol and ribitol. Ribitol feeds into the pentose and glucuronate interconversion pathway from riboflavin metabolism and is a starting point for the conversion of carbohydrates to lipids, amino acids, and starch resources. We hypothesize ribitol depletion indicates corals have high photosynthetic rates and are able to produce compounds

needed to maintain nutrition. At elevated treatment conditions, ribitol accumulates by 125 %, suggesting downstream metabolic processes are not activated, possibly due to limited nutritional resources, which in turn could impede the conversion of carbohydrates. These findings exemplify potential advancements in our mechanistic understanding of coral response to environmental stressors made possible with the application of metabolomic techniques.

4 Conclusions

Our study is the first to comprehensively profile metabolites and lipids in reef-building corals as a function of combinatorial global climate change stressors. While the individual effects of high temperature and ocean acidification on the metabolome cannot be parsed in this study, we are able to demonstrate that corals alter their metabolic pathways to maintain cellular homeostasis during exposure to stressors associated with global climate change. Shifts in pathways involved in carbohydrate metabolism, cell structural maintenance, and cell signaling promotes conservation of energy and fortifies defense mechanisms to maintain cellular activity required for survival, often at the expense of growth and reproduction (Gasch et al. 2000). Metabolites and pathways identified from our approach provide targets for future analyses seeking to develop a comprehensive mechanistic understanding of coral response to disturbance. To determine if corals alter these pathways in response to all stressors, a logical next step is to tease apart the impacts of multiple stressors using fully factorial experimental designs. Our results demonstrate the capacity of metabolomics to detect impacts of global climate change on reefs, and promote the use of metabolite detection techniques across biological systems.

Acknowledgments We thank the National Science Foundation Hawaii Experimental Program to Simulate Competitive Research (EPS-0903833) and the National Park Service George Melendez Wright Global Climate Change Fellowship Program for supporting this research. We would also like to thank UC Davis West coast metabolomics facility their metabolomic and lipidomic core services. We thank D. Horgen and P. Williams for the advice and input.

Funding This study was funded by the National Science Foundation EPSCoR Program (#EPS-0903833) and the National Park Service George Melendez Wright Global Climate Change Fellowship Program

Compliance with ethical standards

Conflict of Interest All authors declare they have no conflict of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

References

Ainsworth, T. D., Thurber, R. V., & Gates, R. D. (2010). The future of coral reefs: A microbial perspective. *Trends in Ecology & Evolution, 25*, 233–240.

Al-Horani, F. A., Al-Moghrabi, S. M., & de Beer, D. (2003). The mechanism of calcification and its relation to photosynthesis and respiration in the scleractinian coral *Galaxea fascicularis*. *Marine Biology, 142*, 419–426.

Allemand, D., Ferrier-Pagès, C., Furla, P., Houlbrèque, F., Puverel, S., Reynaud, S., et al. (2004). Biomineralisation in reef-building corals: From molecular mechanisms to environmental control. *Comptes Renduls Palevol, 3*, 453–467.

Allemand, D., Tambutté, É., Allemand, D., TambuttE, E., Girard, J., Jaubert, J., et al. (1998). Organic matrix synthesis in the scleractinian coral *Stylophora pistillata*: role in biomineralization and potential target of the organotin tributyltin. *Journal of Experimental Biology, 201*, 2001–2009.

Anthony, K. R. N., Kline, D. I., Diaz-Pulido, G., Dove, S., & Hoegh-Guldberg, O. (2008). Ocean acidification causes bleaching and productivity loss in coral reef builders. *Proceedings of the National Academy of Sciences USA, 105*, 17442–17446.

Aslund, M. W., Celejewski, M., Lankadurai, B. P., Simpson, A. J., & Simpson, M. J. (2011). Natural variability and correlations in the metabolic profile of healthy *Eisenia fetida* earthworms observed using ^{1}H NMR metabolomics. *Chemosphere, 8*, 1096–1101.

Baird, A. H., & Marshall, P. A. (2002). Mortality, growth and reproduction in scleractinian corals following bleaching on the Great Barrier Reef. *Marine Ecology Progress Series, 237*, 133–141.

Banaszak, A. T., Barba Santos, M. G., LaJeunesse, T. C., & Lesser, M. P. (2006). The distribution of mycosporine-like amino acids (MAAs) and the phylogenetic identity of symbiotic dinoflagellates in cnidarian hosts from the Mexican Caribbean. *Journal of Experimental Marine Biology and Ecology, 337*, 131–146.

Banaszak, A. T., LaJeunesse, T. C., & Trench, R. K. (2000). The synthesis of mycosporine-like amino acids (MAAs) by cultured, symbiotic dinoflagellates. *Journal of Experimental Marine Biology and Ecology, 249*, 219–233.

Barshis, D. J., Ladner, J. T., Oliver, T. A., Seneca, F. O., Traylor-Knowles, N., & Palumbi, S. R. (2013). Genomic basis for coral resilience to climate change. *Proceedings of the National Academy of Sciences USA, 110*, 1387–1392.

Bolling, C., & Fiehn, O. (2005). Metabolite profiling of *Chlamydomonas reinhardtii* under nutrient deprivation. *Plant Physiology, 139*, 1995–2005.

Bouslimani, A., Porto, C., Rath, C. M., Wang, M., Guo, Y., Gonzalez, A., et al. (2015). Molecular cartography of the human skin surface in 3D. *Proceedings of the National Academy of Sciences USA, 122*, E2120–E2129.

Brown, B. (1997). Coral bleaching: Causes and consequences. *Coral Reefs, 16*, S129–S138.

Bruno, J. F., & Selig, E. R. (2007). Regional decline of coral cover in the Indo-Pacific: Timing, extent, and subregional comparisons. *PLoS One, 2*, e711.

Bundy, J. G., Davey, M. P., & Viant, M. R. (2009). Environmental metabolomics: a critical review and future perspectives. *Metabolomics, 5*, 3–21.

Burriesci, M. S., Raab, T. K., & Pringle, J. R. (2012). Evidence that glucose is the major transferred metabolite in dinoflagellate-cnidarian symbiosis. *Journal of Experimental Marine Biology and Ecology, 215*, 3467–3477.

Coelho, F. J. R. C., Cleary, D. F. R., Rocha, R. J. M., Calado, R., Castanheira, J. M., Rocha, S. M., et al. (2015). Unraveling the interactive effects of climate change and oil contamination on

laboratory-simulated estuarine benthic communities. *Global Change Biology, 21*, 1871–1886.

Constantz, B., & Weiner, S. (1988). Acid macromolecules associated with the mineral phase of scleractinian coral skeletons. *Comparative Biochemistry and Physiology, 248*, 253–258.

Crawley, A., Kline, D., Dunn, S., Anthony, K., & Dove, S. (2010). The effect of ocean acidification on symbiont photorespiration and productivity in *Acropora formosa*. *Global Change Biology, 16*, 851–863.

Davy, S. K., Allemand, D., & Weis, V. M. (2012). Cell biology of cnidarian-dinoflagellate symbiosis. *Microbiology and Molecular Biology Reviews, 76*, 229–261.

De'ath, G., Fabricius, K. E., Sweatman, H., & Puotinen, M. (2012). The 27-year decline of coral cover on the Great Barrier Reef and its causes. *Proceedings of the National Academy of Sciences USA, 109*, 17995–17999.

DeSalvo, M., Sunagawa, S., Voolstra, C., & Medina, M. (2010). Transcriptomic responses to heat stress and bleaching in the elkhorn coral Acropora palmata. *Marine Ecology Progress Series, 402*, 97–113.

Dickinson, G. H., Ivanina, A. V., Matoo, O. B., Pörtner, H. O., Lannig, G., Bock, C., et al. (2012). Interactive effects of salinity and elevated CO_2 levels on juvenile eastern oysters, *Crassostrea virginica*. *Journal of Experimental Marine Biology and Ecology, 215*, 29–43.

Dittami, S. M., Scornet, D., Petit, J.-L., Ségurens, B., Da Silva, C., Corre, E., et al. (2009). Global expression analysis of the brown alga *Ectocarpus siliculosus* (Phaeophyceae) reveals large-scale reprogramming of the transcriptome in response to abiotic stress. *Genome Biology, 10*, R66.

Doney, S. C., Fabry, V. J., Feely, R. A., & Kleypas, J. A. (2009). Ocean acidification: The other CO_2 problem. *Annual Review of Marine Science, 1*, 169–192.

Doney, S. C., Ruckelshaus, M., Duffy, J. E., Barry, J. P., Chan, F., English, C. A., et al. (2012). Climate change impacts on marine ecosystems. *Annual Review of Marine Science, 4*, 11–37.

Downs, C. A., Ostrander, G. K., Rougee, L., Rongo, T., Knutson, S., Williams, D. E., et al. (2012). The use of cellular diagnostics for identifying sub-lethal stress in reef corals. *Ecotoxicology, 21*, 768–782.

Downs, C. A., Woodley, C. M., Richmond, R. H., Lanning, L. L., & Owen, R. (2005). Shifting the paradigm of coral-reef "health" assessment. *Marine Pollution, 51*, 486–494.

Drupp, P., De Carlo, E. H., Machkenzie, F. T., Bienfang, P., & Sabine, C. (2011). Nutrient inputs, phytoplankton response, and CO_2 variations in a semi-enclosed subtropical embayment, Kaneohe bay, Hawaii. *Aquatic Geochemistry, 17*, 473–498.

Dunn, S. R., Thomas, M. C., Nette, G. W., & Dove, S. G. (2012). A lipidomic approach to understanding free fatty acid lipogenesis derived from dissolved inorganic carbon within cnidarian-dinoflagellate symbiosis. *PLoS One, 7*, e46801.

Ellis, R. P., Spicer, J. I., Byrne, J. J., Sommer, U., Viant, M. R., White, D. A., & Widdicombe, S. (2014). ^{1}H-NMR metabolomics reveals contrasting response by male and female mussels exposed to reduced seawater pH, increased temperature, and a pathogen. *Environmental Technology, 48*, 7044–7052.

Fiehn, O. (2001). Combining genomics, metabolome analysis, and biochemical modelling to understand metabolic networks. *Comparative and Functional Genomics, 2*, 155–168.

Fiehn, O., Wohlgemuth, G., & Scholz, M. (2005). Setup and annotation of metabolomic experiments spectrometric metadata. *Data integration in the life sciences SE—18* (pp. 224–239). Berlin: Springer.

Fiehn, O., Wohlgemuth, G., Scholz, M., Kind, T., Lee, D. Y., Lu, Y., et al. (2008). Quality control for plant metabolomics: reporting MSI-compliant studies. *The Plant Journal, 53*, 691–704.

Fitt, W. K., Gates, R. D., Hoegh-Guldberg, O., Bythell, J. C., Jatkar, A., Grottoli, A. G., et al. (2009). Response of two species of Indo-Pacific corals, *Porites cylindrica* and *Stylophora pistillata*, to short-term thermal stress: the host does matter in determining the tolerance of corals to bleaching. *Journal of Experimental Marine Biology and Ecology, 373*, 102–110.

Gardner, T. A., Côté, I. M., Gill, J. A., Grant, A., & Watkinson, A. R. (2003). Long-term region-wide declines in Caribbean corals. *Science, 301*, 958.

Gasch, A. P., Spellman, P. T., Kao, C. M., Carmel-harel, O., Eisen, M. B., Storz, G., et al. (2000). Genomic expression programs in the response of yeast cells to environmental changes. *Molecular Biology of the Cell, 11*, 4241–4257.

Gates, R. D., & Ainsworth, T. D. (2011). The nature and taxonomic composition of coral symbiomes as drivers of performance limits in Scleractinian corals. *Journal of Experimental Marine Biology and Ecology, 408*, 94–101.

Gattuso, J.-P., Allemand, D., & Frankignoulle, M. (1999). Photosynthesis and calcification at cellular, organismal and community levels in coral reefs: a review on interactions and control by carbonate chemistry. *American Zoologist, 39*, 160–183.

Gigon, A., Matos, A.-R., Laffray, D., Zuily-Fodil, Y., & Pham-Thi, A.-T. (2004). Effect of drought stress on lipid metabolism in the leaves of *Arabidopsis thaliana* (ecotype Columbia). *Annals of Botany, 94*, 345–351.

Glynn, P. W. (1984). Widespread coral mortality and the 1982-83 El Nino warming event. *Environmental Conservation, 11*, 133–146.

Gordon, B. R., & Leggat, W. (2010). Symbiodinium—invertebrate symbioses and the role of metabolomics. *Marine Drugs, 8*, 2546–2568.

Gordon, B., Leggat, W., & Motti, C. (2013). Extraction protocol for nontargeted NMR and LC-MS metabolomics-based analysis of hard coral and their algal symbionts. In U. Roessner & D. A. Dias (Eds.), *Metabolomics tools for natural product discovery* (Vol. 1055, pp. 129–147). Dordrecht: Humana Press.

Goreau, T. F., Goreau, N. I., & Yonge, C. M. (1971). Reef corals: Autotrophs or heterotrophs? *Biological Bulletin, 141*, 247–260.

Goreau, T. J., & Macfarlane, A. H. (1990). Reduced growth rate of *Montastrea annularis* following the 1987–1988 coral bleaching event. *Coral Reefs, 8*, 211–215.

Grottoli, A. G., Rodrigues, L. J., & Juarez, C. (2004). Lipids and stable carbon isotopes in two species of Hawaiian corals, *Porites compressa* and *Montipora verrucosa*, following a bleaching event. *Marine Biology, 145*, 621–631.

Guy, C., Kaplan, F., Kopka, J., Selbig, J., & Hincha, D. K. (2008). Metabolomics of temperature stress. *Physiologia Plantarum, 132*, 220–235.

Hammer, K. M., Pedersen, S. A., & Størseth, T. R. (2012). Elevated seawater levels of CO_2 change the metabolic fingerprint of tissues and hemolymph from the green shore crab *Carcinus maenas*. *Comparative Biochemistry and Physiology Part D, 7*, 292–302.

Hoegh-Guldberg, O., Mumby, P. J., Hooten, A. J., Steneck, R. S., Greenfield, P., Gomez, E., et al. (2007). Coral reefs under rapid climate change and ocean acidification. *Science, 318*, 1737.

Hughes, T. P., Rodrigues, M. J., Bellwood, D. R., Ceccarelli, D., Hoegh-Guldberg, O., McCook, L., et al. (2007). Phase shifts, herbivory, and the resilience of coral reefs to climate change. *Current Biology, 17*, 360–365.

IPCC. (2014). Climate Change 2014: Synthesis report. Contribution of working groups 1, II, and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, IPCC, Geneva, Switzerland.

Jury, C. P., Robert, W. F., & Alina, S. M. (2010). Effects of variations in carbonate chemistry on the calcification rates of *Madracis auretenra* (=*Madracis mirabilis* sensu Wells, 1973): bicarbonate

concentrations best predict calcification rates. *Global Change Biology, 16*, 1632–1644.

Kanehisa, M., & Goto, S. (2000). KEGG : Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research, 28*, 27–30.

Kaniewska, P., Campbell, P. R., Kline, D. I., Rodriguez-Lanetty, M., Miller, D. J., Dove, S., & Hoegh-Guldberg, O. (2012). Major cellular and physiological impacts of ocean acidification on a reef building coral. *PLoS One, 7*, e34659.

Kaplan, F., Kopka, J., Haskell, D. W., Zhao, W., Schiller, K. C., Gatzke, N., et al. (2004). Exploring the temperature-stress metabolome. *Plant Physiology, 136*, 4159–4168.

Kaplan, F., Kopka, J., Sung, D. Y., Zhao, W., Popp, M., Porat, R., & Guy, C. L. (2007). Transcript and metabolite profiling during cold acclimation of *Arabidopsis* reveals an intricate relationship of cold-regulated gene expression with modifications in metabolite content. *The Plant Journal, 50*, 967–981.

Kind, T., & Fiehn, O. (2006). Metabolite profiling in blood plasma. *Metabolomics: Methods and protocols* (pp. 3–18). Totowa: Humana Press.

Kind, T., Liu, K., Lee, D. Y., Defelice, B., Meissen, J. K., & Fiehn, O. (2013). LipidBlast in silico tandem mass spectrometry database for lipid identification. *Nature Methods, 10*, 755–758.

Klueter, A., Crandall, J., Archer, F., Teece, M., & Coffroth, M. (2015). Taxonomic and environmental variation of metabolite profiles in marine dinoflagellates of the genus *Symbiodinium*. *Metabolites, 5*, 74–99.

Kopp, C., Domart-Coulon, I., Escrig, S., Humbel, B. M., Hignette, M., & Meibom, A. (2015). Subcellular investigation of photosynthesis-driven carbon assimilation in the symbiotic reef coral *Pocillopora damicornis*. *mBio, 6*, e02299–14.

Krediet, C., Ritchie, K., Paul, V., & Teplitski, M. (2013). Coral-associated micro-organisms and their roles in promoting coral health and thwarting diseases. *Proceedings of the Royal Society B, 280*, 20122328.

Kroeker, K. J., Kordas, R. L., Crim, R. N., & Singh, G. G. (2010). Meta-analysis reveals negative yet variable effects of ocean acidification on marine organisms. *Ecology Letters, 13*, 1419–1434.

Kültz, D. (2005). Molecular and evolutionary basis of the cellular stress response. *Annual Review of Physiology, 67*, 225–257.

Langdon, C., & Atkinson, M. J. (2005). Effect of elevated pCO_2 on photosynthesis and calcification of corals and interactions with seasonal change in temperature/irradiance and nutrient enrichment. *Journal Geophysical Research, 110*, 1–16.

Lardon, I., Eyckmans, M., Vu, T., Laukens, K., Boeck, G., & Dommisse, R. (2013). ^{1}H-NMR study of the metabolome of a moderately hypoxia-tolerant fish, the common carp (*Cyprinus carpio*). *Metabolomics, 9*, 1216–1227.

Lesser, M. P., Mazel, C. H., Gorbunov, M. Y., & Falkowski, P. G. (2004). Discovery of symbiotic nitrogen-fixing cyanobacteria in corals. *Science, 305*, 997–1000.

Lesser, M. P. (2011). Coral bleaching: causes and mechanisms. In Z. Dubinsky & N. Stambler (Eds.), *Coral reefs: An ecosystem in transistion* (pp. 405–419), Springer.

Loya, Y. (2001). Coral bleaching: The winners and the losers. *Ecology Letters, 4*, 122–131.

Marubini, F., Ferrier-Pages, C., Furla, P., & Allemand, D. (2008). Coral calcification responds to seawater acidification: A working hypothesis towards a physiological mechanism. *Coral Reefs, 27*, 491–499.

Matyash, V., Liebisch, G., Kurzchalia, T. V., Shevchenko, A., & Schwudke, D. (2008). Lipid extraction by methyl-tert-butyl ether for high-throughput lipidomics. *Journal of Lipid Research, 49*, 1137–1146.

McHardy, I. H., Goudarzi, M., Tong, M., Ruegger, P. M., Schwager, E., Weger, J. R., et al. (2013). Integrative analysis of the microbiome and metabolome of the human intestinal mucosal surface reveals exquisite inter-relationships. *Microbiome, 1*, 17.

Mendes, J. M., & Woodley, J. D. (2002). Timing of reproduction in *Montastraea annularis*: Relationship to environmental variables. *Marine Ecology Progress Series, 277*, 241–251.

Meyer, E., & Weis, V. M. (2012). Study of cnidarian-algal symbiosis in the "omics" age. *Biological Bulletin, 223*, 44–65.

Michal, G., & Schomburg, D. (Eds.). (1999). *Biochemical Pathways: An atlas of biochemistry and molecular biology*. New York: Wiley.

Moberg, F., & Folke, C. (1999). Ecological goods and services of coral reef ecosystems. *Ecological Economics, 29*, 215–233.

Moya, A., Huisman, L., Ball, E. E., Hayward, D. C., Grasso, L. C., Chua, C. M., et al. (2012). Whole transcriptome analysis of the coral *Acropora millepora* reveals complex responses to CO_2-driven acidification during the initiation of calcification. *Molecular Ecology, 21*, 2440–2454.

Muscatine, L. (1967). Glycerol excretion by symbiotic algae from corals and *Tridacna* and its control by the host. *Science, 156*, 516–519.

Muscatine, L., & Cernichiari, E. (1969). Assimilation of photosynthetic products of zooxanthellae by a reef coral. *The Biological Bulletin, 137*, 506–523.

Muscatine, L., & Porter, J. W. (1977). Reef corals: mutualistic symbioses adapted to nutrient-poor environments. *BioScience, 27*, 454–460.

Oku, H., Yahiro, H., & Onaga, K. (2003). Lipid biosynthesis from [^{14}C]-glucose in the coral *Montipora digitata*. *Fisheries Science, 69*, 625–631.

Pandolfi, J. M., Connolly, S. R., Marshall, D. J., & Cohen, A. L. (2011). Projecting coral reef futures under global warming and ocean acidification. *Science, 333*, 418–422.

Papina, M., Meziane, T., & van Woesik, R. (2003). Symbiotic zooxanthellae provide the host-coral *Montipora digitata* with polyunsaturated fatty acids. *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology, 135*, 533–537.

Parsons, H. M., Ekman, D. R., Collette, T. W., & Viant, M. R. (2009). Spectral relative standard deviation: A practical benchmark in metabolomics. *Analyst, 134*, 478–485.

Patton, J. S., Abraham, S., & Benson, A. A. (1977). Lipogenesis in the intact coral *Pocillopora capitata* and its isolated zooxanthellae: Evidence for a light-driven carbon cycle between symbiont and host. *Marine Biology, 44*, 235–247.

Peng, S. E., Chen, C. S., & Song, Y. F. (2012). Assessment of metabolic modulation in free-living versus endosymbiotic *Symbiodinium* using synchrotron radiation-based infrared microspectroscopy. *Biology Letters, 8*, 434–437.

Putnam, H., & Gates, R. D. (2015). Preconditioning the reef-buildling coral *Pocillopora damicornis* and the potenital for trans-generation acclimatization in coral larvae under future climate change conditions. *Journal of Experimental Marine Biology and Ecology, 218*, 2365–2372.

Putron, S., McCorkle, D., Cohen, A., & Dillon, A. (2011). The impact of seawater saturation state and bicarbonate ion concentration on calcification by new recruits of two Atlantic corals. *Coral Reefs, 30*, 321–328.

Rahav, O., Dubinsky, Z., Achituv, Y., & Falkowski, P. G. (1989). Ammonium metabolism in the zooxanthellate coral, *Stylophora pistillata*. *Proceedings of the Royal Society of London B: Biological Sciences, 236*, 325–337.

Raina, J., Tapiolas, D., Willis, B. L., & Bourne, D. G. (2009). Coral-associated bacteria and their role in the biogeochemical cycling of sulfur. *Applied and Environmental Microbiology, 75*, 3492–3501.

Ramos-Silva, P., Kaandorp, J., Herbst, F., Plasseraud, L., Alcaraz, G., Stern, C., et al. (2014). The skeleton of the staghorn coral *Acropora millepora*: Molecular and structural characterization. *PLoS One, 9*, e97454.

Ries, J. B., Cohen, A. L., & McCorkle, D. C. (2009). Marine calcifiers exhibit mixed responses to CO_2-induced ocean acidification. *Geology, 37*, 1131–1134.

Rodolfo-Metalpa, R., Martin, S., Ferrier-Pages, C., & Gattuso, J.-P. (2010). Response of the temperate coral *Cladocora caespitosa* to mid- and long-term explosure to pCO_2 and temperature levels projected for the year 2100 AD. *Biogeosciences, 7*, 289–300.

Rosenberg, E., Koren, O., Reshef, L., Efrony, R., & Zilber-Rosenberg, I. (2007). The role of microorganisms in coral health, disease and evolution. *Nature Reviews Microbiology, 5*, 355–362.

Rowher, F., Serguritan, V., Azam, F., & Knowlton, N. (2002). Diversity and distribution of coral-associated bacteria. *Marine Ecology Progress Series, 243*, 1–10.

Schock, T. B., Stancyk, D. A., Thibodeaux, L., Burnett, K. G., Burnett, L. E., Boroujerdi, A. F., & Bearden, D. W. (2010). Metabolomic analysis of Atlantic blue crab, *Callinectes sapidus*, hemolymph following oxidative stress. *Metabolomics, 6*, 250–262.

Seneca, F. O., & Palumbi, S. R. (2015). The role of transcriptome resilience in resistance of corals to bleaching. *Molecular Ecology, 7*, 1467–1484.

Shankar, V., Homer, D., Rigsbee, L., Khamis, H. J., Michail, S., Raymer, M., et al. (2015). The networks of human gut microbe-metabolite associations are different between health and irritable bowel syndrome. *The ISME Journal, 9*, 1899–1903.

Shinzato, C., Shoguchi, E., Kawashima, T., Hamada, M., Hisata, K., Tanaka, M., et al. (2011). Using the *Acropora digitifera* genome to understand coral responses to environmental change. *Nature, 476*, 320–323.

Sogin, E. M., Anderson, P., Williams, P., Chen, C.-S., & Gates, R. D. (2014). Application of ^{1}H-NMR metabolomic profiling for reef-building corals. *PLoS One, 9*, e111274.

Stitt, M., Lunn, J., & Usadel, B. (2010). *Arabidopsis* and primary photosynthetic metabolism—more than the icing on the cake. *The Plant Journal, 61*, 1067–1091.

Tapiolas, D. M., Motti, C., Holloway, P., & Boyle, S. G. (2010). High levels of acrylate in the Great Barrier Reef coral *Acropora millepora*. *Coral Reefs, 29*, 621–625.

Tarrant, A. M., Atkinson, S., & Atkinson, M. J. (1999). Estrone and estradiol-17β concentration in tissue of the scleractinian coral, *Montipora verrucosa*. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology, 122*, 85–92.

Tarrant, A. M., Blomquist, C. H., Lima, P. H., Atkinson, M. J., & Atkinson, S. (2003). Metabolism of estrogens and androgens by scleractinian corals. *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology, 136*, 473–485.

Thurber, R. L., Barott, K. L., Hall, D., Liu, H., Rodriguez-Mueller, B., Desnues, C., et al. (2008). Metagenomic analysis indicates that stressors induce production of herpes-like viruses in the coral *Porites compressa*. *Proceedings of the National Academy of Sciences USA, 105*, 18413–18418.

Trygg, J., & Wold, S. (2002). Orthogonal projections to latent structures (O-PLS). *Journal of Chemometrics, 16*, 119–128.

van Den Berg, R. A., Hoefsloot, H. C. J., Westerhuis, J. A., Smilde, A. K., & Van Der Werf, M. (2006). Centering, scaling, and transformations: Improving the biological information content of metabolomics data. *BMC Genomics, 7*, 142.

van Meer, G., Voelker, D. R., & Feigenson, G. W. (2008). Membrane lipids: Where they are and how they behave. *Nature Reviews Molecular Cell Biology, 9*, 112–124.

Viant, M. R., Werner, I., Rosenblum, E. S., Gantner, A. S., Tjeerdema, R. S., & Johnson, M. L. (2003). Correlation between heat-shock protein induction and reduced metabolic condition in juvenile steelhead trout (*Oncorhynchus mykiss*) chronically exposed to elevated temperature. *Fish Physiology and Biochemistry, 29*, 159–171.

von Holt, C. (1968). Uptake of glycine and release of nucleoside-polyphosphates by zooxanthellae. *Comparative Biochemistry and Physiology, 3*, 1071–1079.

Wagner, N. D., Hillebrand, H., Wacker, A., & Frost, P. C. (2013). Nutritional indicators and their uses in ecology. *Ecology Letters, 16*, 535–544. doi:10.1111/ele.12067.

Whitehead, L. F., & Douglas, A. E. (2003). Metabolite comparisons and the identity of nutrients translocated from symbiotic algae to an animal host. *Journal of Experimental Biology, 206*, 3149–3157.

Wikoff, W. R., Anfora, A. T., Liu, J., Schultz, P. G., Lesley, S. A., Peters, E. C., & Siuzdak, G. (2009). Metabolomics analysis reveals large effects of gut microflora on mammalian blood metabolites. *Proceedings of the National Academy of Sciences USA, 106*, 3698–3703.

Yancey, P. H. (2005). Organic osmolytes as compatible, metabolic and counteracting cytoprotectants in high osmolarity and other stresses. *Journal of Experimental Biology, 208*, 2819–2830.

Yancey, P., Heppenstall, M., & Ly, S. (2010). Betaines and dimethylsulfoniopropionate as major osmolytes in cnidaria with endosymbiotic dinoflagellates. *Physiological and Biochemical Zoology, 83*, 167–173.

Yellowlees, D., Rees, T. A. V., & Leggat, W. (2008). Metabolic interactions between algal symbionts and invertebrate hosts. *Plant, Cell and Environment, 31*, 679–694.

Yost, D. M., Wang, L.-H., Fan, T.-Y., Chen, C.-S., Lee, R. W., Sogin, E., & Gates, R. D. (2013). Diversity in skeletal architecture influences biological heterogeneity and *Symbiodinium* habitat in corals. *Zoology (Jena), 116*, 262–269.

Zhang, P. (1993). Model selection via multifold cross validation. *The Annals of Statistics, 21*, 299–313.